

LDV_H2

Hydrogen Fuel Cell Light Duty Vehicle



Hydrogen Car

Inputs / Outputs

dir	comm	unit	group
Input	H2	PJ	i
Output	PLDVHWY	MPK _m	o
Output	PLDVCTY	MPK _m	o

Efficiency (ceff)

	comm	use2cact	cact2cout	share.up
2	PLDVHWY	1156.069	2.03	0.4
3	PLDVCTY	853.971	2.03	0.8

Key Parameters

cap2act	12.04
olife	16 years
stock	0 1000 Vehicles
cap unit	1000 Vehicles
act unit	million km, city
cost unit	USD/(Vh*km)

Fuel Efficiency

Mode	Fuel	L/100 km	km/kWh	MJ/km	mpg (US)
Highway	H2	0.72	4.16	0.86	326.31
City	H2	0.98	3.07	1.17	241.04

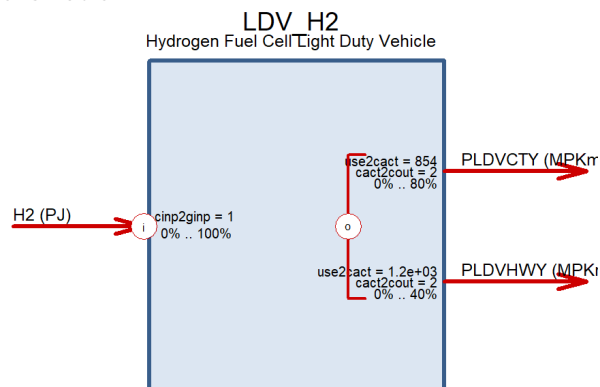
Vehicle Parameters

Fuels	H2	Primary fuel	NA
Efficiency (highway)	0.865 MJ/km	Efficiency (city)	1.171 MJ/km
Annual distance	12,043 km/yr	Capacity unit	1000 vehicles
Load/vehicle (hwy)	2.03 passengers	Load/vehicle (city)	2.03 passengers
Max share (highway)	40%	Max share (city)	80%
Vehicle cost	58,916 USD	Annual fixed cost	13 USD/yr
Lifetime	16 yr		

Capacity bounds

region	year	stock	cap.lo	cap.up	cap.fx	ncap.lo	ncap.up	ncap.fx	ret.lo	ret.up	ret.fx
IND	2024	0	NA	NA	NA	NA	NA	NA	NA	NA	NA

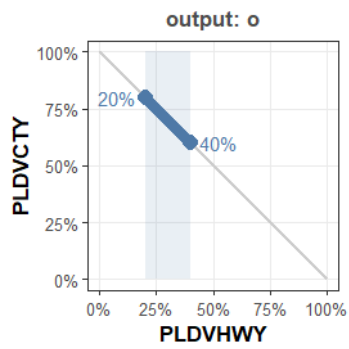
Schematic



Costs [USD/(Vh*km)]

Fixed O&M	0.01
Investment cost	58.92

Share Mix Constraints



Levelized Cost 0.68 [USD/(Vh*km)], 0.33–0.34 [USD/(Passenger*km)]

